

Guanyu Zhang

✉ evanz1627@gmail.com

🌐 guanyu-zhang.github.io

🌐 [guanyuzhang](https://www.linkedin.com/in/guanyuzhang)

🌐 [guanyu-zhang](https://github.com/guanyu-zhang)

EDUCATION

Columbia University

M.S. in Computer Science, Machine Learning Track (Course Assistant: COMS 6156 Cloud Computing)

New York, NY

Dec 2022

The University of Texas at Austin

B.S. in Mathematics (Honors: College Scholars)

Austin, TX

Aug 2021

WORK EXPERIENCE

Ant International (Ant Group)

Sunnyvale, CA

Software Engineer

Jan 2024 – Present

- Built and deployed an AI-powered testing framework using **RAG** and a **LoRA**-tuned Qwen3 model, automating end-to-end test case generation from product requirement documents through structured retrieval and distilled code-test training data.
- Built a production-grade in-memory cache framework in **Java**, decoupling legacy DAO dependencies for reusability and adding **transaction** support for atomic cache operations, supporting **over \$10B GMV** and **2M daily transactions**.
- Developed a native web-search tool in **Python** for a multi-agent system, deploying **metasearch engines** and **crawler servers** with **Kubernetes** to improve latency performance and reduce tool API failure rate.
- Optimized Amazon compliance APIs using **Kafka** to handle asynchronous database writes and **Airflow** and **Cassandra** for data aggregation and storage, reducing response latency by **50%** and increasing API availability under peak load.
- Executed tenant-level database migration from **Alicloud** to **Google Cloud Platform** using **Jenkins**, **Flink**, and **Kafka**, automating data synchronization to ensure consistency and minimize downtime during cutover.

Alipay US, Inc. (Ant Group)

Sunnyvale, CA

Software Engineer

Jan 2023 – Dec 2023

- Implemented a cross-border bank account service using **Java Spring Boot** with **distributed transaction processing**, designing **idempotent** and **retry-safe** transaction handling to ensure consistency for over **\$12B** annual transfer volume across **15+** regions.
- Engineered a payment orchestration layer for Venmo and PayPal integration on AliExpress, scaling a **Kubernetes**-based 500-pod microservice system to handle **3,000–5,000 QPS** and serve millions of daily buyers with consistent reliability.
- Rebuilt Alipay's Shopify payment integration by replacing legacy SDKs with new API-based payment interfaces, launching wallet and card payment apps with **Redis** caching and **asynchronous message queues**, meeting Shopify's integration requirements and serving over **50,000** global merchants with high availability.
- Implemented a **CI/CD** pipeline with **Jenkins** for the digital wallet frontend using **JavaScript**, automating build and deployment workflows to improve release efficiency and reduce manual interventions.

Qingdao Anzhi Capital Management

Shanghai, China

Software Engineer Internship

Oct 2020 – Jan 2021

- Built and maintained a cloud-based trading data pipeline on **AWS**, using **RDS** and **DynamoDB** to support reliable ingestion, storage, and retrieval of daily market data.
- Engineered alpha factors from multi-resolution **OHLC** market data for both long-term market forecasting and short-term index-option prediction, and developed models with **scikit-learn** and **PyTorch**.
- Evaluated models, **visualized** predictive signals, and exposed results through an internal API for the quantitative-trading team.

PROJECTS

Open-Source Contributions

Sep 2025 – Present

- Contributed to **OpenClaw** by improving dependency handling and environment isolation for **containerized** AI agents.
- Contributed to **Lobe Hub** by developing agent tooling in its open-source AI framework written in **TypeScript**.

Columbia University COMS 4113 Distributed Systems KV Cache Project

Jan 2022 – May 2022

- Built a **key-value cache** in **Go** with **concurrent access control**, **TTL-based eviction**, and **snapshot-based recovery**, focusing on correctness, fault tolerance, and state consistency.

Speech Data Annotation Reliability Project

Nov 2020 – Dec 2020

- Developed a pipeline using **MFCC** features and pre-trained **ResNet** models to process raw speech data, and built a **VGG-S** classifier that improved **F1 score by 5%** over the RF baseline.

TECHNICAL SKILLS

C++, Go, Java, Python, JavaScript, SQL; Docker, Kubernetes; AWS & GCP; MySQL, MongoDB, Redis; PyTorch, scikit-learn, Transformers, PySpark